

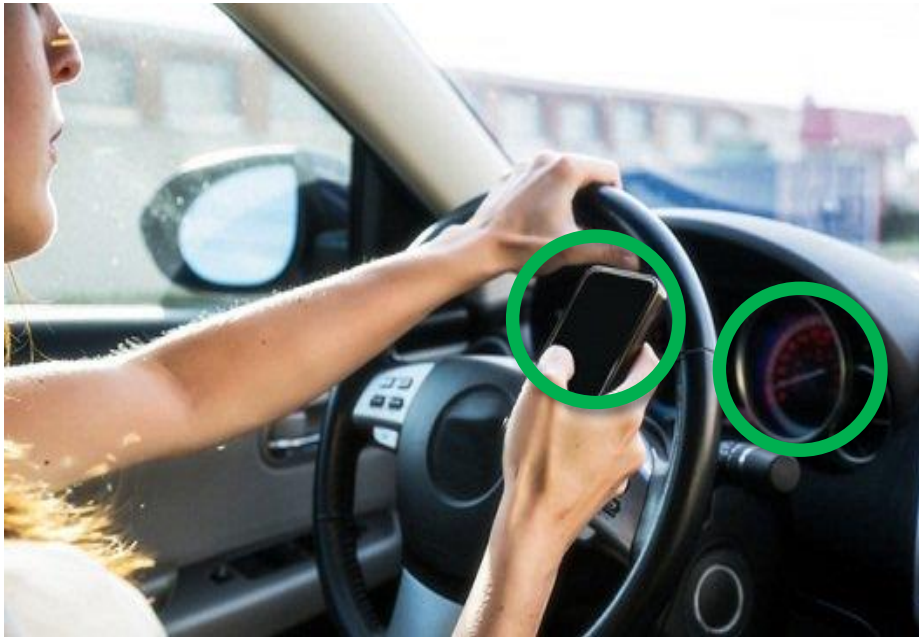


new perspective on way

Crystal Vision



Prepared by
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Nowadays people are «bound» to their phones. It is strictly prohibited to use the phone without a handset while driving in most of the countries. Prohibition doesn't solve the problem though cause people go on chatting using SMS or 'messengers'. The number of accidents on the road keeps up growing. After the ban came out in California this number has increased by 12%.



About 78% of people use the smartphone behind the wheel



Looking away from the road for just 2 seconds increases the risk of an accident by 9 to 24-fold



Only eye refocusing takes 0.3 -0.6 sec which gives extra 50 feet to braking distance



Reflex time increases by 400% while typing SMS or checking email at the wheel



One in four car accidents is caused by cell phone use while driving



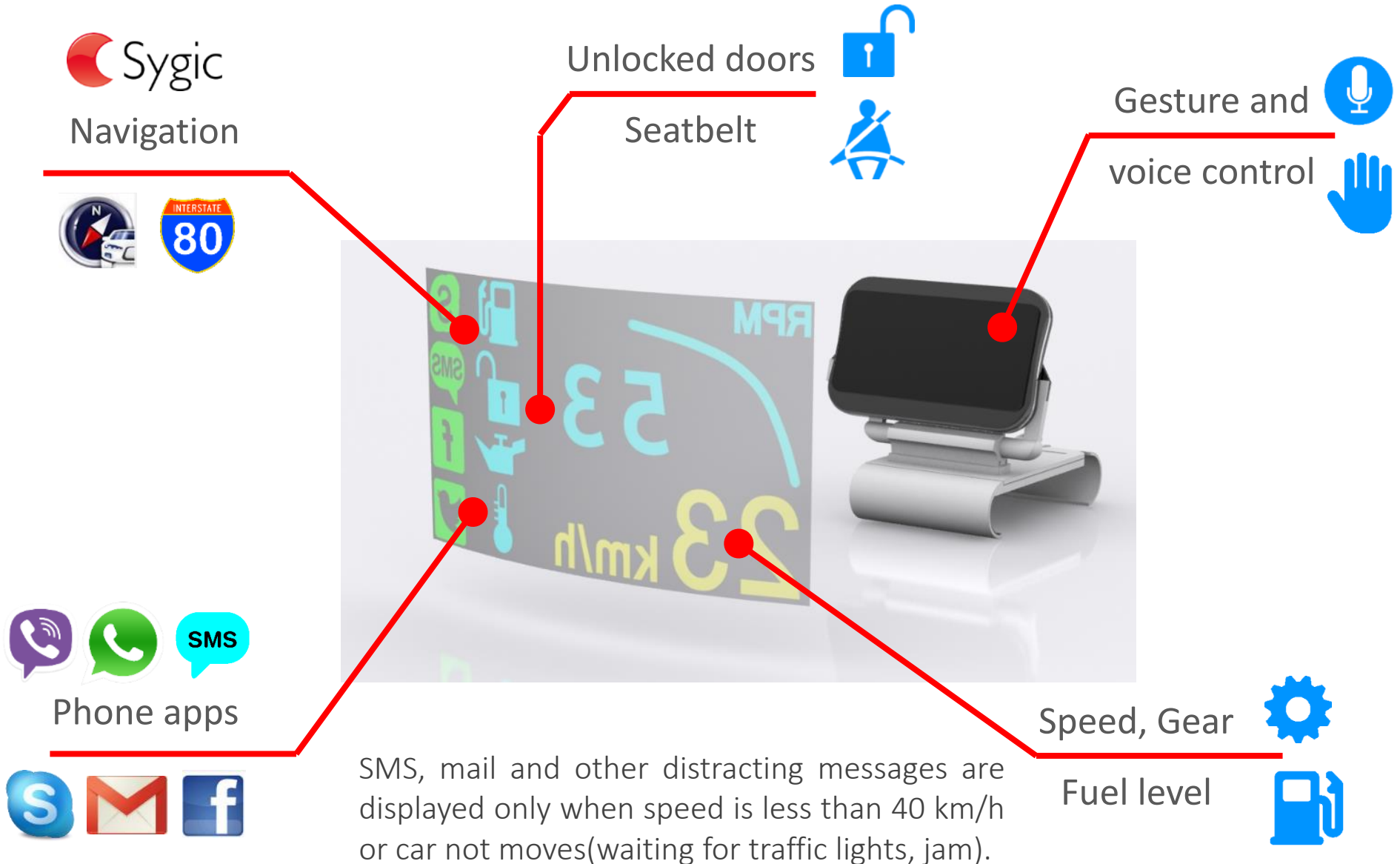
270,000 people die in the world per year due to the mobile phone use



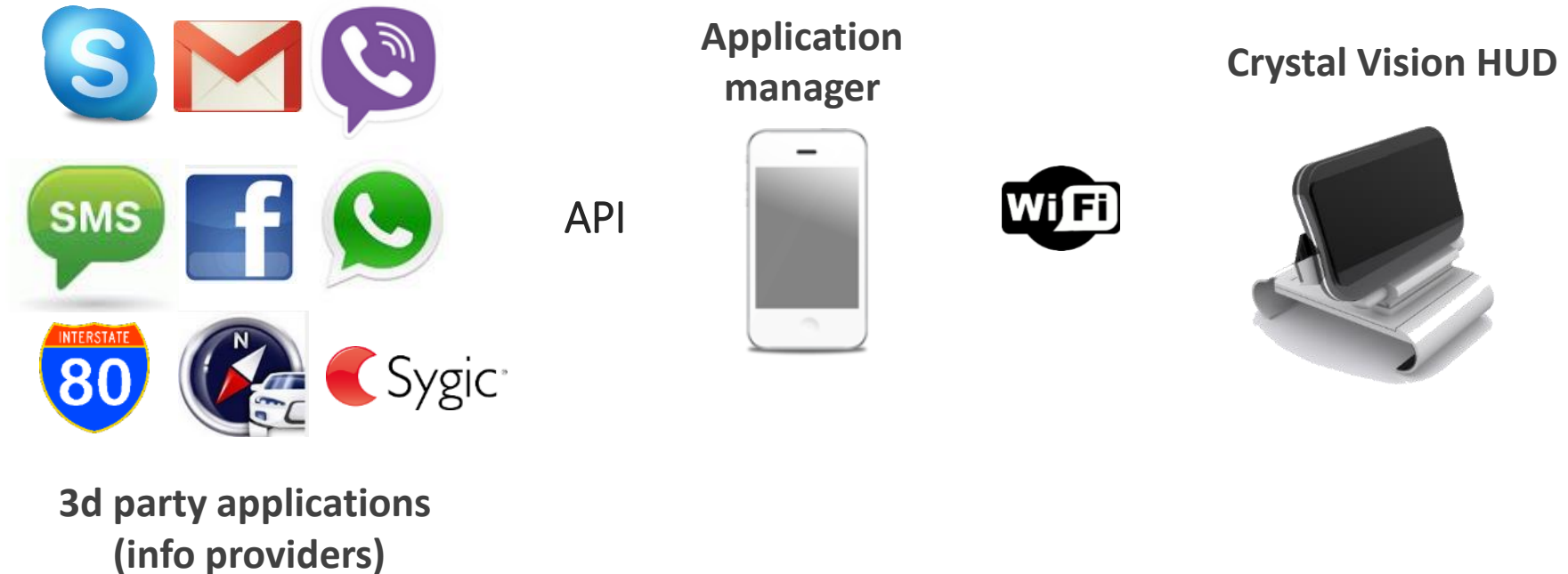
3 million people are hospitalized each year because of the phone usage while driving.

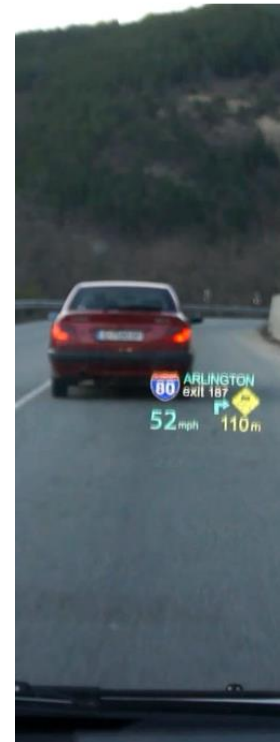


**World Health
Organization**



Crystal Vision is a portable HUD system, that projects image on a windshield of your car. Driver and passengers see it as a semitransparent image that “levitates” above the car bumper and is not seen from outside. Projector is controlled via an app installed on the smartphone. The application aggregates data from the cloud services and smartphone applications, onboard computer and sends it in a special format to the projector. [See the video.](#)







Currently in the World there are only a few competitors that offer embedded HUD systems. They are early bird companies that only offer pre-ordering their products.

Continental provides factory installed solutions for luxury cars segment. Audi, BMW, Mercedes: all those brands are equipped with Continental HUD systems if option is chosen when ordering car.



Crystal vs. factory



- 👍 Crystal creates pristine image that is of the same quality as created by HUD systems in luxury cars.
- 👍 Crystal HUD is portable and can be installed into any car.
- 👍 Crystal HUD allows a variety of information to be displayed, which is chosen by a user. In contrast, factory HUDs display info is determined by manufacturer, typically, navigation only.
- 👍 Crystal HUD cost is 5 to 10 times less than factory HUD systems and is the same for embedded HUD systems.

- 👍 Since we have solved image-doubling problem, Crystal HUD doesn't require neither reflective plates nor films, which are used in other competitive products. Image is created right on a windshield.
- 👍 Crystal HUD can be tuned up to show image below, on or above horizon line. Image is well seen in any conditions: day and night, sun and rain, snow and fog. Image is also seen by passengers.
- 👍 Image created by Crystal HUD is remote, it "levitates" over the car hood and appears to be 1,7 – 2,3 m up the driver's eyes. This allows not to refocus the sight as the image is already in the field of vision. Driver doesn't get distracted from the road. According to the latest research, that is **crucial point for driving safety** and differs Crystal HUD from other pseudo "HUD" systems.

Crystal vs. portable



**Competitive
portable "HUD"**

Look at the video, comparing factory HUD and Crystal Vision HUD

Factory HUD

Corvette 5C
GM HUD Bezel
Monochrome image
Price **\$1100**



Crystal Vision

Opel Zafira
Crystal Vision HUD
Colorful image
Price **\$250**



Total Available Market (TAM)

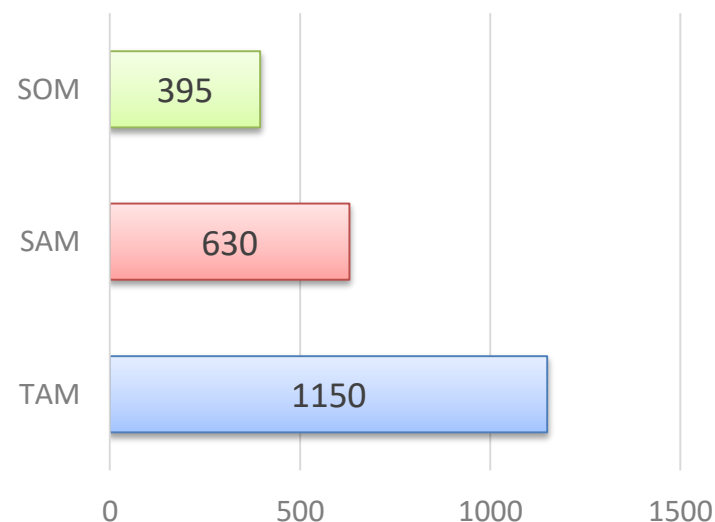
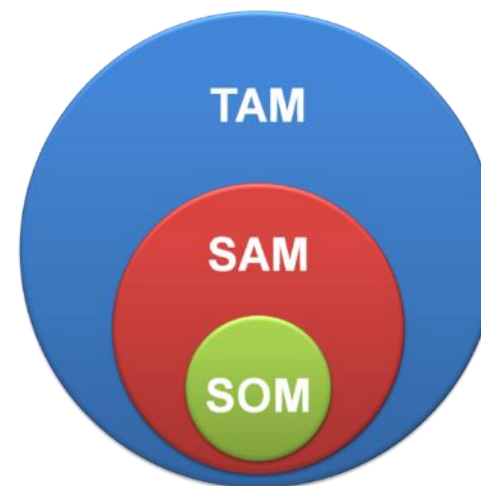
= 1150 m. – all cars in the world
driven every day

Serviceable Available Market (SAM) = TAM

x 87% - passenger cars
x 90% - cars that support technology
x 70% - use smartphone while driving
= 630 million

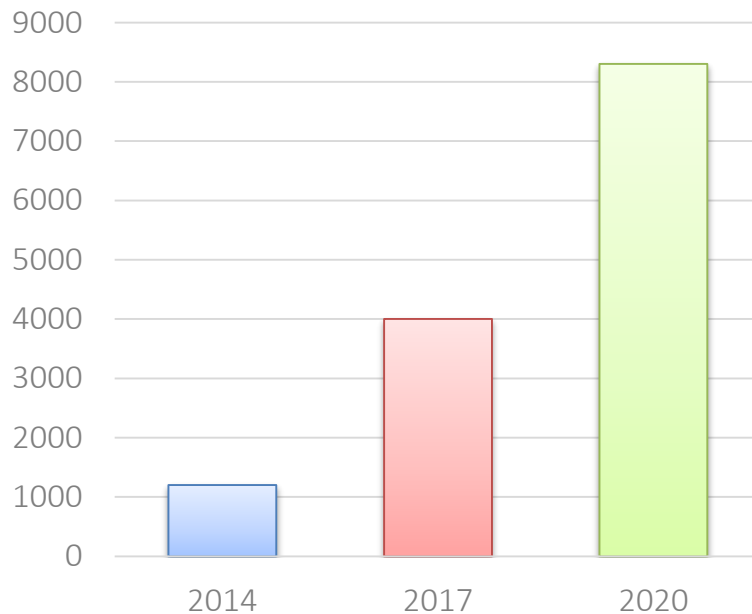
Serviceable Obtainable Market (SOM) =

Russia 40 m. passenger cars +
China 117 m. passenger cars +
USA 220 m. passenger cars +
Europe 250 m. passenger cars
= 627 x 90% x 70% = 395 million

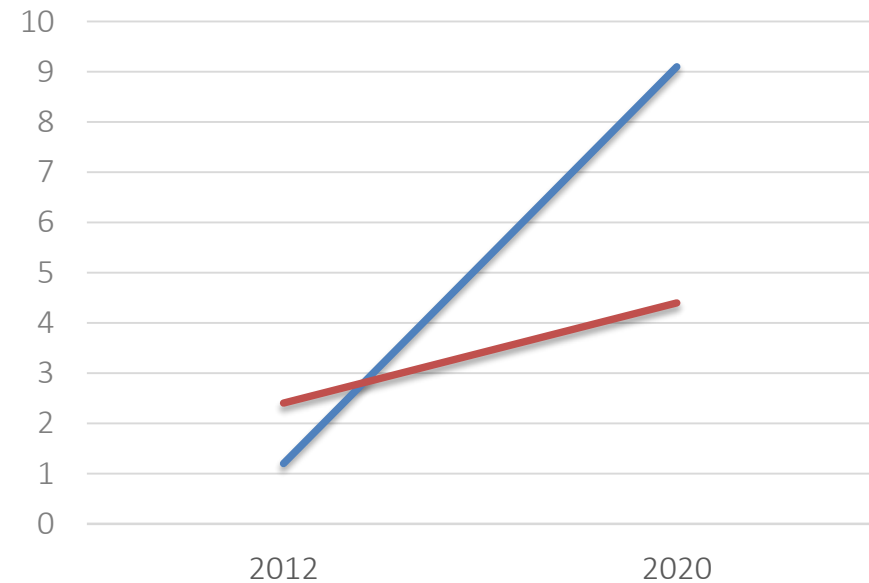


According to a new market research report, the global head-up display market is estimated to reach USD 9.02 Billion by 2020, at a CAGR of 26.23% between 2015 and 2020. The market for the automotive application is expected to grow at a CAGR of 29.28% during the forecast period. This growth is driven by the increasing demand for mid-segment and sport cars globally.

(source: MarketsandMarkets)



Fast growing market,
millions of USD



— Growth of number of cars with HUD option,
millions



- Online sales and retail
- Massive sales via crowdfunding platforms like Kickstarter
- Subscription on required Crystal applications and services or freemium model for our services.
- Ads broadcast while car not moves and on parking



- Wholesales via distributor chains
- Revenue-sharing with application developers for using Crystal Vision API
- Product customization for auto producers (OEM, White Labeling).
- License manufacturing and installation for auto producers (we have patented design)

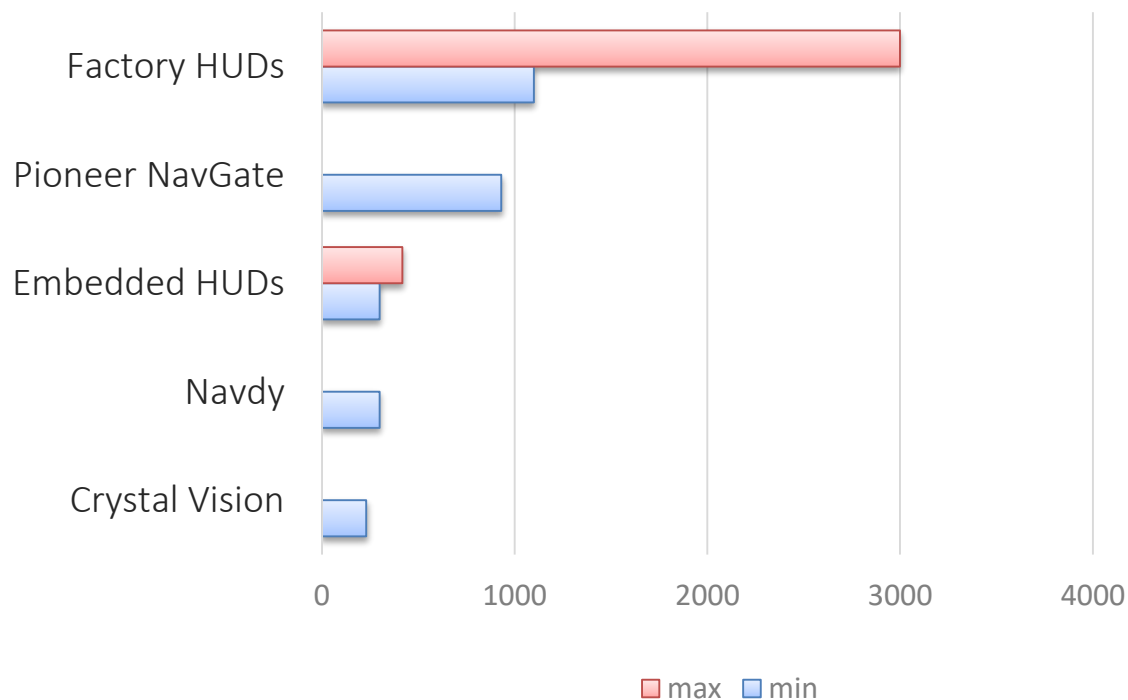
Who is interested in the product:

- Our audience - car owners.
- Car producers.

How to promote the product:

- Actually, distributors from all over the world already find us and offer partnership, distribution and promotion in their countries.
- Sales web-page <http://crystal-hud.com>.
- Attractive car stickers and shields in the set with the device.
- The device advertises itself attracting drivers and passengers' attention with its outer look (from mouth to mouth marketing or "the rumour mill").
- Equipment of foreign car rent services chains (Top Rent Car, Herz, etc.) or test-drive cars will let many people get to know the product.
- Commercials on web-sites devoted to cars (auto.ru, autoblog.com, auto.de).
- Gadgets overview web-sites (digitaltrends.com, cnet.com, engadget.com).
- Car magazines, brochures, TV-shows and video casts ("Top gear", "Big test-drive"), social network communities, messages in blogs.

- Factory HUD system \$1100 - \$3000
- Pioneer HUD system NavGate \$930
- NAVDY pre-order \$299 (\$499)
- Crystal Vision HUD \$250



Crystal Vision projector

1. Patent pending (Russian) 100%
2. Patent pending (international) 35%
3. Trademark registration 70%
4. Minimum Functional Prototype (MFP) 100%
5. Complete Functional prototype (CFP) 92%
 - optic elements 100%
 - electronics (PCB) 85%
 - enclosure and mount 90%
6. Design-For-Manufacture(DFM) 59%
 - optic elements 100%
 - prototype elements 90%
 - molds 40%
 - certification 5%
7. Suppliers of electronic components, elements and materials 100 %
8. Potential resources for manufacture, assembly, packaging 100 %
9. Packaging 0%
10. First factory run (FFR) 0%
11. Retail 0%



Software development

1. Crystal Vision Application Manager
 - for Android 85%
 - for iOS 10%
2. Remote animations engine “Remote Canvas”
 - for Android 95%
 - for iOS 10%

Common

1. Kickstarter campaign 70%
2. Distribution channels
 - USA 40%
 - China 0%
 - Finland 50%
 - Germany 10%
 - Thailand 70%
 - Bulgaria 70%
3. Marketing 2%



Estimated progress status at the end of 2015 - 1Q 2016**Crystal Vision projector**

1. Patent pending (Russian) 100%
2. Patent pending (international) 80%
3. Trademark registration 100%
4. Minimum Functional Prototype (MFP) 100%
5. Complete Functional prototype (CFP) 100%
6. Design-For-Manufacture (DFM) 80 - 100%
7. Suppliers of electronic components, elements and materials 100%
8. Potential resources for manufacture, assembly, packaging 100%
9. Packaging 100%
10. First factory run (FFR) 60 - 100%
11. Retail 60 - 90%

Software development

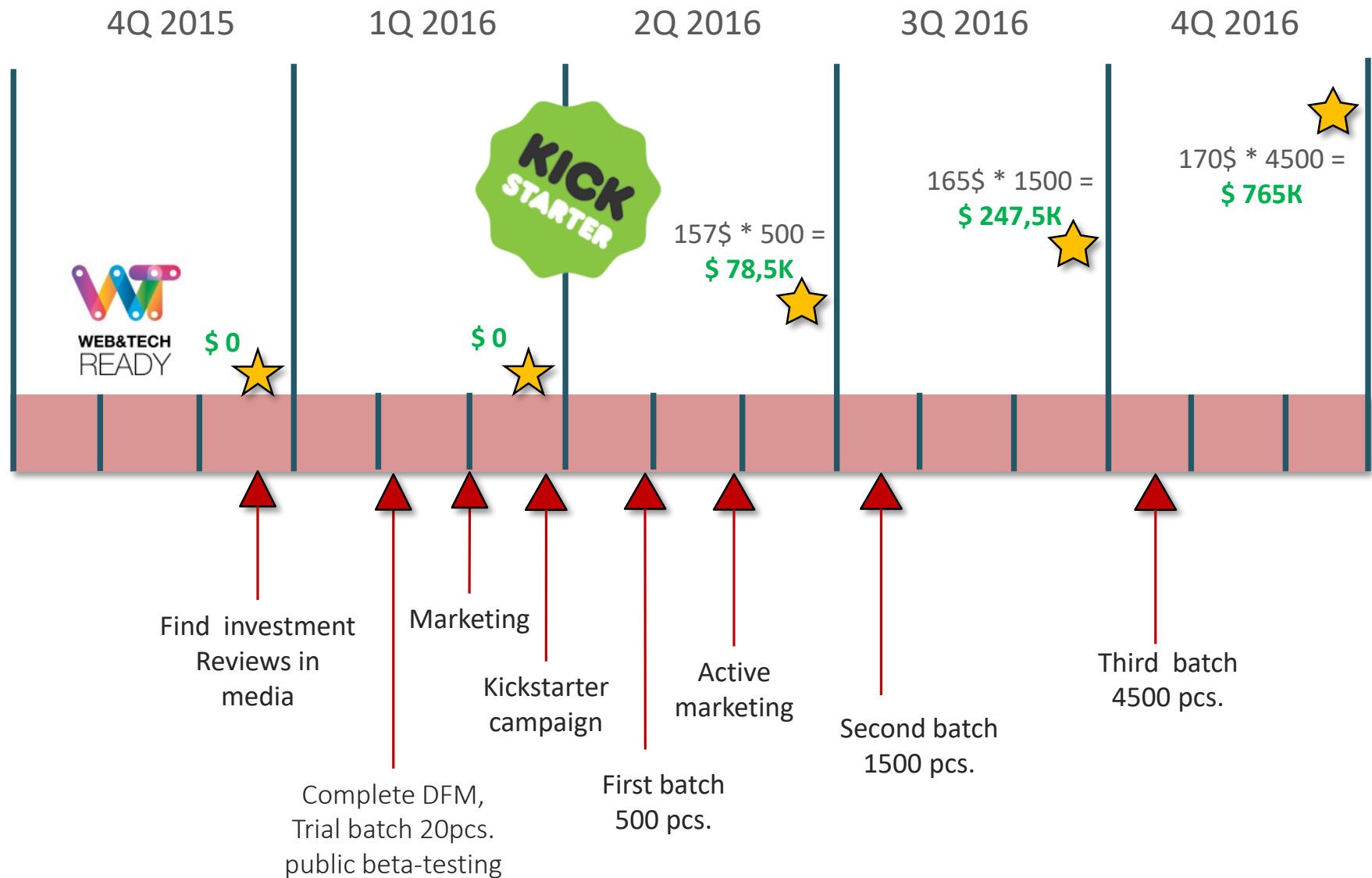
1. Crystal Vision application manager 100%
2. Remote animations engine "Remote Canvas" 100%

Common

1. Kickstarter campaign 100%
2. Distribution channels 90%
3. Marketing 60-100%



- **Fixed production costs \$65K**
 - Industrial design finalization \$21K
 - hardware finalization \$11K
 - software finalization \$10K
 - Assembly \$32K
 - Packaging \$6K
 - Engineering Validation Test(EVT) \$2K
 - Certification \$2K
 - Design Verification Test(DVT) \$2K
- **Delivery \$12 per unit**
- **Variable costs \$16 per unit**
- **Business trips 6 months \$10K**
- **Outsource and custom works \$5K**
- **International Patent finalization \$20K**
- **First 5K batch (KickStarter campaign, distributor channels)**
 - Incomes - \$1,250,000(5K x \$250)
 - Fixed costs \$65 per unit
 - Profit per unit \$157
(\$250 - \$65 - \$12 - \$16)
 - Profit for batch \$785,000
(5K x \$157)
- **Second and another batches profit per unit \$170**
- **Distribution channels and Wholesale (FOB price \$180) profit per unit \$95
(\$180 - \$65 - \$4 Delivery - \$16)**



#	Risk	Description	Threat	Remediation
1	Economic	Foreign exchange volatility influences the purchasing ability in different countries and price setting of the product.	High	Price division according to the country. We made a survey in different countries what price the customers are ready to pay for it.
2	Production	Nonfulfillment of baseline work, failure to meet the deadline.	Moderate	We already have partners with two-time stock volumes. We are setting up new contacts with partners in different countries for product sales.
3	Legal	Prohibition on Crystal HUD implemented systems usage in certain countries.	Low	Crystal HUD doesn't use any cover-up on windshield, no reflection glass which can disturb the driver. Switch to other markets.
4	Low demand	Low demand	Moderate, Low	Percent order in relation to the overall number of clicks on the site, 19%. Presentation, survey in the US, 65%. Presentation, survey in Germany, 71%.
5	Strategic	Patents infringement, manufacturing of products functionally close to it.	High	Intellectual property rights, protection of interests in court.
6	Compliance	Ban of Application Manager on AppStore and Google Play Market.	Low	Compliance with the rules and guidelines for application development.
7	Legal, Regulatory	Ecological norms and rules violation, problems of standardizing and licenses.	Low	We don't use any toxic, poisoning or contaminating substances. We use mercury-free and lead-free components and soldering, RoHS compliant.

Until recently we didn't participate in any events,
but when we began we won several prizes:



We are the member of Google Launchpad program.

We won top second place in the most prestigious Russian competition. There were 712 competing projects from 14 countries.



We won high technology acceleration program in Skolkovo Innovation Center.

We've patented technology in RosPatent. And now still pending for technology enhancement patent.





We conducted equipping cars with Crystal Vision HUD, demonstration to drivers in the United States, Germany, Bulgaria, Greece, Spain, Finland, China, Thailand, Russia and others.

We surveyed several hundreds of drivers, and about 65% of them showed an enthusiastic interest in the Crystal Vision HUD and the desire to buy it.



We spent about 3 years for technology R&D, and now have finished it. Product is 75-80% ready and we will start production soon.

We need money, for:

- Finalizing industrial prototype
- Preparation for mass production
- First batch production
- Marketing
- Preparing for Kickstarter campaign (commercial and reviews in media)
- Outsourced works (design, software)
- Patenting, certification and other legal affairs

We have of about 220 preorders now. Genuine interest to our product, distribution offers, crowdfunding success of related projects ([Navdy](#), [HUDWAY](#), [Expolride](#)) and our competitive advantages let us assume great success.



Let's turn every car into a safe spaceship,
and make business together

Any questions :

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Analytics, statistics, development prospects

- <http://www.marketsandmarkets.com/Market-Reports/head-up-display-hud-market-684.html>
- <http://press.ihs.com/press-release/design-supply-chain/automotive-head-display-market-goes-high-gear>
- <http://www.acea.be>
- https://en.wikipedia.org/wiki/Motor_vehicle
- <http://chartsbin.com/view/1114>
- https://en.wikipedia.org/wiki/List_of_countries_by_vehicles_per_capita
- <http://www.who.int/en/>

HUD products

- <http://www.youtube.com/watch?v=-8SP1OSIpaM>
- <http://www.edmunds.com/mazda/mazda3/2014/long-term-road-test/2014-mazda-mazda3-heads-up-display.html>
- <http://www.pioneer.eu/eur/products/archive/SPX-HUD01/page.html>
- <http://www.youtube.com/watch?v=Qg8dq9FIX78>